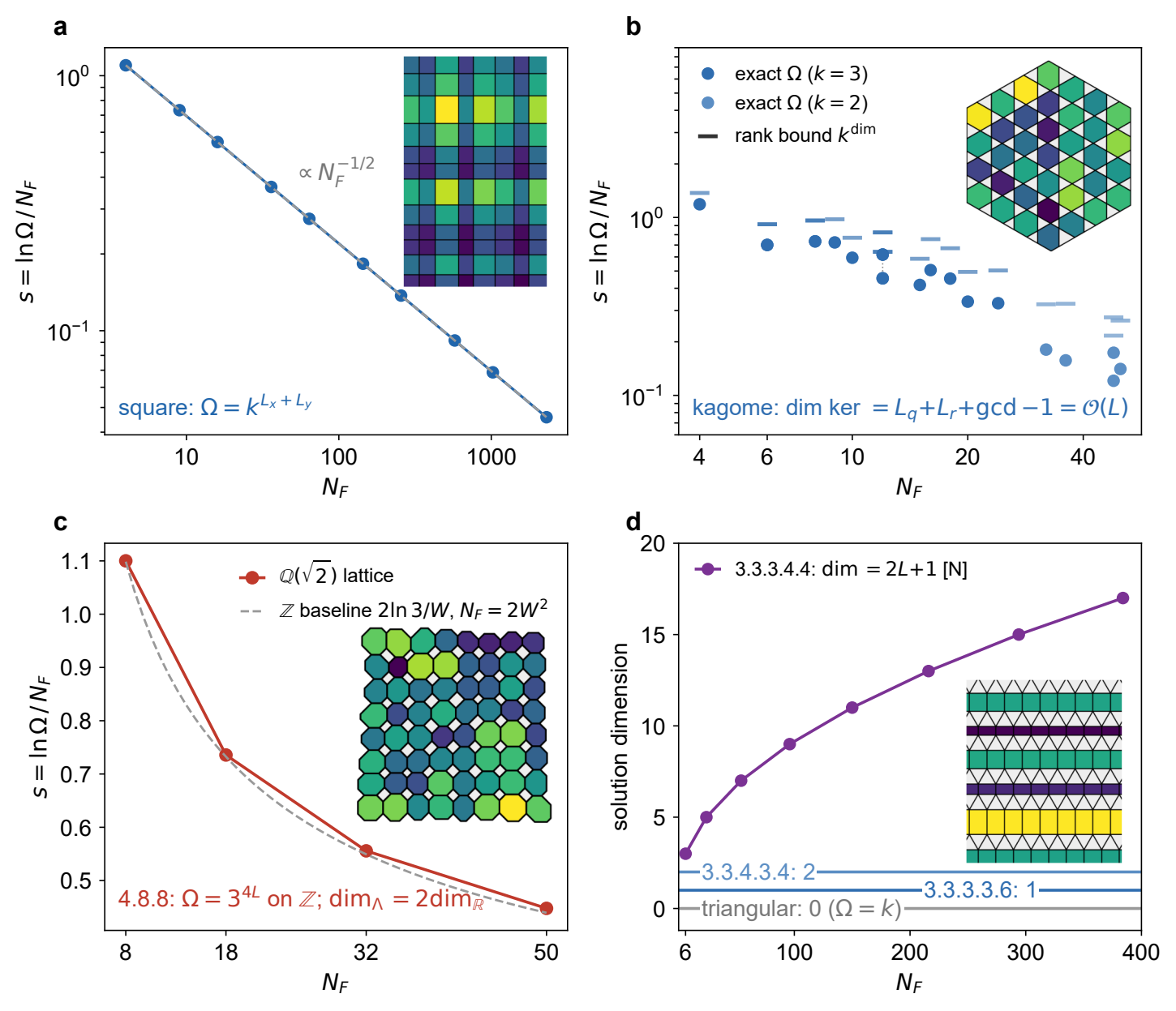

a
10^0
10^{-1}
$s = \ln\Omega / N_F$
$\propto N_F^{-1/2}$
square: $\Omega = k^{L_x + L_y}$
10
100
1000
N_F
b
exact Ω ($k = 3$)
exact Ω ($k = 2$)
rank bound $k^{\dim}$
10^0
10^{-1}
$s = \ln\Omega / N_F$
kagome: dim ker $= L_q + L_r + \gcd - 1 = \mathcal{O}(L)$
4
6
10
20
40
N_F
c
$\mathbb{Q}(\sqrt{2})$ lattice
$\mathbb{Z}$ baseline $2\ln 3/W$, $N_F = 2W^2$
1.1
1.0
0.9
0.8
0.7
0.6
0.5
$s = \ln\Omega / N_F$
4.8.8: $\Omega = 3^{4L}$ on $\mathbb{Z}$; $\dim_\Lambda = 2\dim_{\mathbb{R}}$
8
18
32
50
N_F
d
3.3.3.4.4: dim $= 2L+1$ [N]
20
15
10
5
0
solution dimension
3.3.4.3.4: 2
3.3.3.3.6: 1
triangular: 0 ($\Omega = k$)
6
100
200
300
400
N_F